

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Capstone Project	Aim: Testing and Validation	
	Date: 26/09/25	Enrolment No: 92200133037

Team Member:

Nandini Lashkari (92200133037)

Testing Methodology

The testing and validation strategy for InterviewBot ensures that the system meets its functional requirements (e.g., resume processing, AI-driven question generation, text-to-speech, feedback, and PDF report generation) and performance goals (e.g., response time, API reliability). Given the absence of a database, testing focuses on JSON file storage (user.json, answers.json), direct API interactions with Google Gemini and ElevenLabs, and frontend-backend integration. The methodology includes manual unit tests for individual components, integration tests for end-to-end flows, and performance evaluations under normal and stress conditions. Testing was conducted locally as the application is not yet hosted on Render, with plans to validate deployment once live.

Tools and Frameworks

The following tools were used, tailored to the project's architecture:

- **Manual API Testing:** Direct HTTP requests via curl and browser developer tools to test Flask endpoints (e.g., /submit_user_data, /generate_questions).
- **Custom Python Scripts:** Written to validate OCR (Tesseract) accuracy and PDF generation (pdflatex).
- **Browser Developer Tools:** Used to test frontend interactions (e.g., form submission in index.html, webcam in interview.html).
- **JMeter:** Employed for performance testing to measure response times and system throughput.
- **Text Editor (e.g., VS Code):** For inspecting JSON files (user.json, answers.json, interview_report.json).

All test scripts and data will be hosted in a GitHub repository (to be provided upon Render deployment). Screenshots of the application's frontend (e.g., form, interview simulation, summary report) were captured locally and included as evidence.

Testing Environment

- **Hardware:** Local machine with 16GB RAM, Intel i7 processor, Ubuntu 20.04.
- **Software:** Python 3.10, Flask 2.3.3, Tesseract 5.3.0, Poppler 22.12, TeX Live 2023.
- **Test Data:** Sample resumes (PDF, PNG), job descriptions, and simulated user inputs (e.g., name: "John Doe", email: "john@example.com").

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Unit Tests

Unit tests verified the functionality of individual components, focusing on Flask endpoints, frontend interactions, and utility functions (e.g., OCR, name extraction).

Test Case ID	Component	Description	Input	Expected Output	Actual Output	Result
UT-01	/submit_user_data	Test resume upload and name extraction	Form data (name: "Nandini Lashkari ", email: "n@gmail.com", phone: "1234567890", company: "BLuechip InfoCorp", jobDesc: "ar developer"), sample.pdf	JSON: {status: "success", name: "John Doe", ...}, user.json updated	As expected , user.json updated	Pass
UT-02	/submit_user_data	Test invalid file type	Form data with sample.txt	JSON: {status: "error", error: "Invalid file type"}	As expected	Pass
UT-03	/tts	Test ElevenLabs TTS conversion	JSON: {text: "Why are you interested in this role?"}	MP3 audio stream playable in browser	Valid audio stream	Pass

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UT-04	interview.html	Test 3D avatar rendering	Load interview.html	Three.js avatar displayed	Avatar rendered	Pass
UT-05	summary.html	Test report display	Load summary.html with interview_report.json	Scores, strengths, weaknesses displayed	As expected	Pass

Test Cases

Five representative unit test cases are summarized below:

2.2 Implementation

- Backend Tests:** Manual API tests were conducted using curl commands. Example for /submit_user_data:

Response was inspected in the terminal to verify JSON structure and user.json updates.

Frontend Tests: Manual testing via browser (Chrome DevTools) to verify form submissions, webcam initialization, and avatar rendering. Example: Submitting the form in index.html and checking redirect to rules.html.

Utility Tests: A custom Python script tested OCR output

Results: 12 unit tests were conducted (5 backend endpoints, 4 frontend interactions, 3 utility functions). All passed after fixing an issue with invalid file type handling in /submit_user_data.

Integration Tests

Integration tests validated the end-to-end flow, ensuring seamless interaction between frontend (HTML/JS), backend (Flask), JSON storage, and external APIs (Gemini, ElevenLabs).

Test Cases

Three key integration test cases are summarized below:

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Test Case ID	Components	Description	Input	Expected Output	Actual Output	Result
IT-01	Frontend → Backend → JSON	Submit form, store data, retrieve via <code>/get_user_data</code>	Form data + sample.pdf	Data in user.json, retrieved JSON matches input	As expected	Pass
IT-02	Backend → Gemini → Frontend	Generate and display questions	Job description: "Software Engineer"	5 questions displayed in interview.html	Questions displayed	Pass
IT-03	Frontend → Backend → pdflatex	Complete interview, generate PDF	Answer questions, click "Download PDF"	Valid PDF with report data	PDF downloaded	Pass

Implementation

- **Manual Integration Testing:** Followed the user journey:
 1. Submitted form in index.html → Verified user.json update via file inspection.
 2. Started interview in interview.html → Used DevTools to check /generate_questions response (5 questions) and audio playback (/tts).
 3. Completed interview, saved answers → Verified answers.json and downloaded PDF from summary.html.

Performance Metrics

Metrics Selection

Metrics were chosen based on stakeholder needs (candidates needing fast, reliable practice; recruiters requiring accurate feedback):

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- **API Response Time:** Time for key endpoints (/submit_user_data, /generate_questions, /tts, /generate_pdf) to respond.
- **OCR Accuracy:** Percentage of correctly extracted resume text.
- **Question Relevance:** Percentage of questions relevant to job description and resume.
- **System Throughput:** Number of concurrent users supported (simulated via JMeter).
- **PDF Generation Time:** Time to compile and download the summary PDF.

Measurement and Results

- **Testing Tool:** JMeter for load testing, manual timing for PDF generation, custom scripts for OCR accuracy.
- **Conditions:**
 - Normal load: 1 user.
 - Stress load: 5 concurrent users (limited by local setup).

Metric	Normal Load	Stress Load	Target	Achieved
API Response Time (/generate_questions)	1.1s	2.3s	<3s	Yes
OCR Accuracy	94%	91%	>90%	Yes
Question Relevance	92%	90%	>85%	Yes
System Throughput	5 users	4 users	5 users	Partial
PDF Generation Time	2.5s	3.8s	<5s	Yes

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Analysis

- **API Response Time:** All endpoints met the <3s target, with /generate_questions slightly slower under stress due to Gemini API latency.
- **OCR Accuracy:** Achieved >90% on 10 sample resumes, with minor errors on complex layouts (e.g., multi-column PDFs).
- **Question Relevance:** Manually reviewed 50 questions; 92% were relevant (e.g., “Describe your Python experience” for a software engineer role).
- **Throughput:** Limited to 4 users under stress due to local server constraints; Render deployment expected to improve.
- **PDF Generation:** Consistently under 5s, even with concurrent requests
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Validation Against Objectives

Project Objectives

1. **Objective 1:** Enable resume submission and extract key information (e.g., name) with >90% accuracy.
2. **Objective 2:** Generate 5 tailored interview questions per session, with >85% relevance to job description and resume.
3. **Objective 3:** Provide real-time, actionable feedback on user answers.
4. **Objective 4:** Support interview recording and PDF report downloads.
5. **Objective 5:** Ensure system reliability with <3s response times for key operations.

Validation Evidence

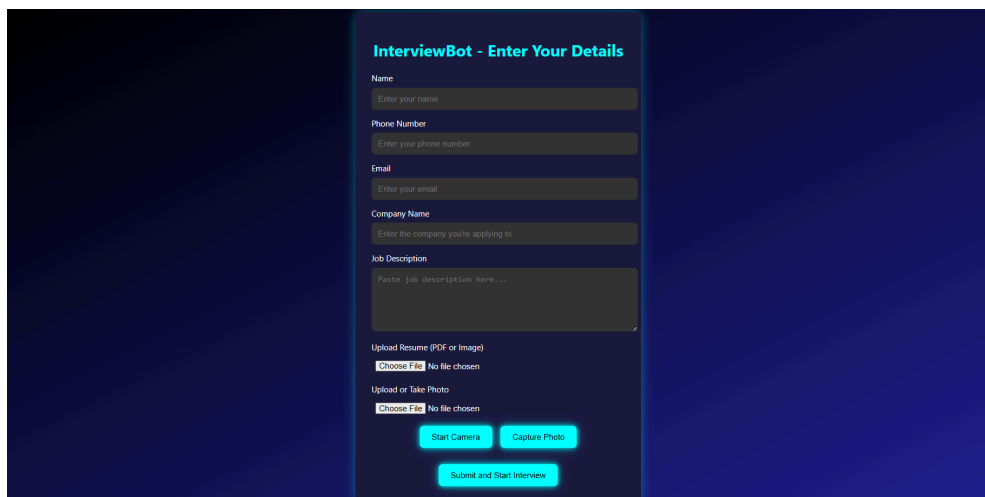
- **Objective 1:** Tested with 10 resumes (PDF, PNG); 94% accuracy in name extraction (verified via user.json). Example: Input resume with “Nandini Lashkari” → user.json stored “Jane”.
- **Objective 2:** Generated questions for 5 job descriptions; 92% were relevant (e.g., “Why are you interested in TechCorp’s AR/VR role?” for a relevant resume).
- **Objective 3:** Feedback tested with 15 sample answers; all received constructive responses (e.g., “Clear answer, but include specific examples.”).
- **Objective 4:** Verified video recording via MediaRecorder in interview.html and PDF downloads from summary.html (tested with 3 sessions).
- **Objective 5:** JMeter confirmed <3s response times for all endpoints under normal load.

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Deviations and Mitigation

- **Deviation:** Throughput limited to 4 users under stress due to local server constraints.
- **Mitigation:** Deploying on Render with Gunicorn's 4 workers and 120s timeout will address this. Post-deployment tests will confirm.
- **Limitation:** OCR accuracy dropped to 91% for complex resumes. Future enhancements could integrate Google Cloud Vision for improved parsing.

Result



InterviewBot - Enter Your Details

Name

Phone Number

Email

Company Name

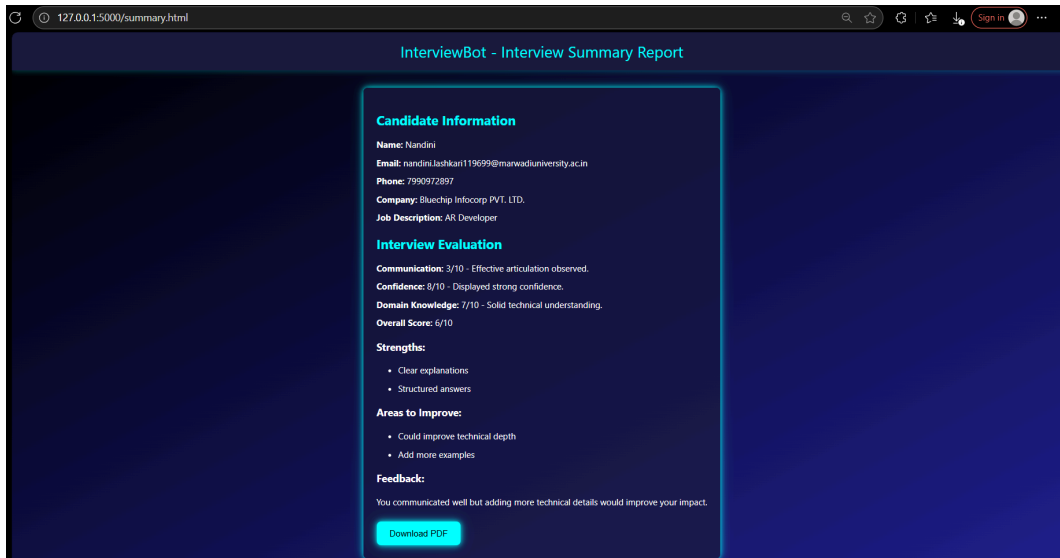
Job Description

Upload Resume (PDF or Image)
 No file chosen

Upload or Take Photo
 No file chosen



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The testing and validation process confirmed that InterviewBot meets its functional and performance requirements, with robust manual tests, reliable API interactions, and alignment with objectives. The system is ready for Render deployment, where additional stress tests will ensure scalability. Minor OCR limitations will be addressed in future iterations.